
PH 211: The Physics Behind Computers

Course Code	PH 211
Course Name	The Physics Behind Computers
L-T-P-C	3
One-time approval	recommended by the 9th BoA (Oct 2015)

The purpose of this course is to attract students into a physics course within a specific context—the fascination Of how computers work, explained from the very basic principles. This course will be really useful to students, because it provides real working examples of the physics they are learning in a relevant, unified context. Students will realize that much of what physicists know about the world goes into the physics of making/ understanding of computers.

Course Contents

- **Introduction:**
 - The Physics Behind Computers
 - Physics, Silicon, and the “Magic” Behind the ‘computer’ Age.
 - A Zoomed-In Look Inside a Computer.
 - Timeline of Great Discoveries and Inventions in Physics and Computer.
- **Mathematics:**
 - The Language of Science and Technology
 - The Utility of Mathematics in Science and Technology.
 - Precision and Significant Digits -Real-World Example, Precision of Display Pixels Large and Small Numbers and Scientific Notation.
 - Units for Physical Quantities
 - Metric-System Units, Binary Numbers. Converting from Decimal to Binary. Converting from Binary to Decimal. Real-World Example -Analog and Digital Variables
 - The Concept of Information
 - Bits, Bytes and Other Units,Exponential Growth
 - Social Impacts: The Exponential Change of Nearly Everything.
- **Mechanics:**
 - Energy Enables Information Technology
 - From Looms to Computers.
 - Speed, Acceleration, and Force.
 - Description of Motion- Distance Travelled, Seek Time of a Hard Drive Head.
 - Force Vectors Net Force Vectors. Electron Acceleration In Cathode Ray Tubes.
 - Principles of Mechanics, Force on a Hard-Drive Head, Gravity, Physics of Energy
 - Friction and Heat Energy, The Constancy of Energy. Motion Sensors in Laptops.
- **Electricity and Magnetism: The Workhorses of Information Technology**
 - Electricity and Magnetism are the Basis of Computers.
 - Electric Charge Electric Forces: Coulomb’s Law, Electric Fields— Field Lines.
 - Electric Current and Conductors Electrical Energy and Voltage.
 - Voltage Sources - Batteries, Energy Stored in a Battery.
 - Energy Storage in a Capacitor, Capacitor Computer Memory.
 - Resistors Conductors, and Ohm’s Law, Electrical Power.
 - Magnetism,Electromagnetism, Electric Current Creates Magnetic Field.
 - The Telegraph, Precursor to the Internet.
 - Changing Electric Field Creates Magnetic Field.

- Changing Magnetic Field Creates Magnetic Field.
- Magnetic Materials and Data Storage.
- **Digital Electronics and Computer Logic:**
 - The Reasoning Abilities of Computers.
 - Concepts of Logic, Electronic Logic Circuits Logic Operations and Diagrams
 - Three-input Logic Operations Building Logic Operations Using the NOR Operation.
 - Using Logic to Perform Arithmetic.
 - Implementing Logic -Electromagnetic Switches. Boolean Search of Databases.
- **Quantum Physics of Atoms and Materials:**
 - Atoms, Crystals and Computers.
 - The Quantum Nature of Electrons and Atoms, Electron Scattering from Crystals.
 - The Spinning of Electrons, Fluorescent Lamps.
 - Electrical Properties of Materials, Conductors, Insulators, Semiconductors.
 - Origin of the Energy Gap in Silicon Crystals.
 - Atomic Nature of Magnetic Domains.
- **Semiconductor Physics:**
 - Transistors and Circuits Silicon, Transistors and Computers
 - Controlling the Conductivity of Silicon p-n Junctions and Diodes.
 - Rectifying an alternating signal, A Simple Crystal AM Radio Receiver
 - Transistors, CMOS Computer Logic, “Water-Effect Transistors.
 - Miniaturization, Integrated Circuits, and Photolithography.
 - Silicon Crystal Preparation Lithography for Fabricating a p-n Junction.
 - Bipolar Transistors.
- **Digital Memory and Computers:**
 - Physics, Memory and Computers.
 - Sequential Logic for Computer Memory.
 - The Set-Reset Latch, Enabled Data Latch or D-Latch.
 - Static Random-Access Memory.
 - SRAM with Six Transistors.
 - Dynamic Random-Access Memory, Nonvolatile Memory. Quantum Tunneling.
 - Magnetic Tape and Hard Disk Memory, Optical Compact Disk Memory.
 - Error Immunity of Digital Data, The Structure of a Computer.
 - Hierarchy of Computer Memory, Heat-Imposed Limits of Computers
 - Representing Information in Computers using Codes, ASCII de.
 - Coding Images, Data Compression.

Text Books

1. Michael G. Raymer, **Silicon Web-Physics for Internet Age.**

References

1. Paul Hewitt, **Conceptual Physics.**
2. Louis Bloomfield, **How Things Work: The Physics of Everyday Life.**
3. Art Hobson, **Physics: Concepts and Connections.**